

Homework 01

Submission Notices:

- Conduct your homework by filling answers into the placeholders given in this file (in Microsoft Word format). Questions are shown in black color, *instructions/hints are shown in italic and blue color*, and *your content should use any color that is different from those*.
- After completing your homework, prepare the file for submission by exporting the Word file (filled with answers) to a PDF file, whose filename follows the following format,
 <StudentID-1>_<StudentID-2>_HW01.pdf (Student IDs are sorted in ascending order)
 E.g., 2252001_2252002_HW01.pdf
and then submit the file to Moodle directly WITHOUT any kinds of compression (.zip, .rar, .tar, etc.).
- Note that you will get zero credit for any careless mistake, including, but not limited to, the following things.
 1. Wrong file/filename format, e.g., not a pdf file, use “-” instead of “_” for separators, etc.
 2. Disorder format of problems and answers
 3. *Conducted not in English.*
 4. Cheating, i.e., copying other students’ works or let the other student(s) copy your work.

Problem 1. (1pt) Please classify these AI innovations, including *Facebook face detection/recognition*, *Google translate*, *AlphaGo Zero (Automatic Chess Playing)*, *Vehicle counting using traffic surveillance cameras*, *Siri/Alexa/Google Assistant*, *Machine algorithms for automatic heroes in Dota2* into the three given trending field of AI.

- Computer Vision (CV)
- Natural Language Processing (NLP)
- Game Informatics (GI)

Please write your answer in the table

Computer Vision	- Facebook face detection/recognition. - Vehicle counting using traffic surveillance cameras.
Natural Language Processing	- Google translate. - Siri/Alexa/Google Assistant.
Game Informatics	- AlphaGo Zero (Automatic Chess Playing) - Machine algorithms for automatic heroes in Dota2.

Problem 2. (1pt) Discuss that, to what extent, the following systems are instances of AI. That is, you first decide its level of intelligence, low, fair/medium, or high, and then give your reasons.

- **Product 01:** Magnetic keys applied at the parking lot of a university.
- **Product 02:** Temperature and humidity sensors to control automatic watering.
- **Product 03:** Iris recognition for entrance permission at a laboratory.
- **Product 04:** Music composition that inspires musicians with personalized patterns of melodies and genres.

Each product gives 0.25 pt.

Product	Level of intelligence	Reason
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Product 01	Low	Magnetic keys used in parking lots typically operate on a simple mechanism of detecting the presence of a specific magnetic signal to grant access. This system does not exhibit any adaptive or learning capabilities. It merely responds to a predefined input (the presence of a valid magnetic key) with a predefined output (granting access).
Product 02	Fair/Medium	While this system doesn't demonstrate advanced reasoning or decision-making abilities, it does exhibit a certain level of intelligence by dynamically adjusting watering based on real-time environmental conditions (temperature and humidity). It utilizes sensor data to make decisions about watering frequency and duration, adapting to changes in the environment. However, it lacks learning capabilities and operates based on predefined rules or algorithms.
Product 03	Fair/Medium	Iris recognition systems involve some degree of intelligence in identifying individuals based on unique iris patterns. These systems utilize complex algorithms and biometric data processing techniques to match iris patterns with stored templates. While they demonstrate a moderate level of intelligence in terms of pattern recognition and authentication, they operate within predefined parameters and do not adapt or learn over time.
Product 04	High	This product exhibits a high level of intelligence as it involves music composition tailored to individual preferences. Such a system likely utilizes sophisticated AI algorithms, such as machine learning and generative models, to analyze musical tastes and create personalized compositions. It not only requires understanding of musical theory but also adapts and learns from user feedback to refine its output over time. This level of creativity and adaptability suggests a high degree of intelligence in the system.

Problem 3. (1.5pts) *The wolf, goat, and cabbage problem.* Once upon a time a farmer went to a market and purchased a wolf, a goat, and a cabbage. On his way home, the farmer came to the bank of a river and rented a boat. But crossing the river by boat, the farmer could carry only himself and one of his purchases: the wolf, the goat, or the cabbage. If left unattended together, the wolf would eat the goat, or the goat would eat the cabbage. The farmer's challenge was to carry himself and his purchases to the far bank of the river, leaving each purchase intact.

Formulate the above problem as a search problem by answering the following questions.

- How do you represent a state? That is, which elements are included in a single state and what is the range of value for each element?

- State: Location of the farmer, the wolf, the goat, and the cabbage whether they are on the first bank or the second bank. (0 for the first or initial of the bank's river, 1 for another side).

- Elements are in a single state:

- The farmer's location: Can be either 0 or 1.
- The wolf's location: Can be either 0 or 1.
- The goat's location: Can be either 0 or 1.
- The cabbage's location: Can be either 0 or 1.

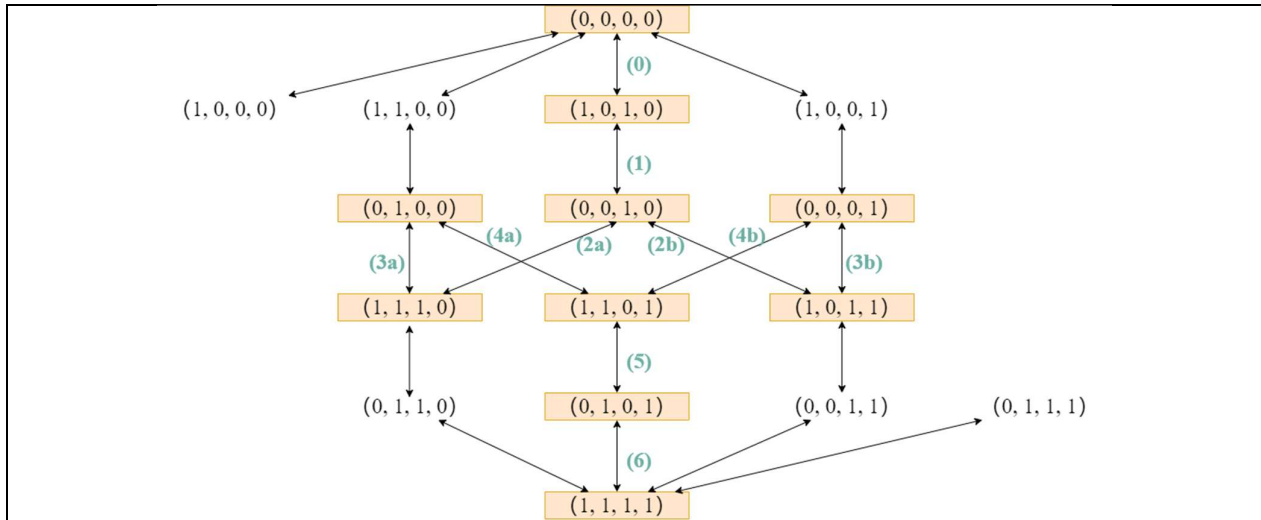
- The range of value for each element:

- The farmer's location : {0, 1}
- The wolf's location: {0, 1}
- The goat's location : {0, 1}
- The cabbage's location: {0, 1}

- How many states are there in state space? Justify your answer.

- There are 4 elements (Location of the farmer, the wolf, the goat, and the cabbage). Each element can be either 0 or 1 so that the maximum states in state space is $2^4 = 16$ states.

- Draw a directed acyclic graph (DAG) of the state space. Mark the optimal solution.



Problem 4. (2pts) Describe the task environment properties for the game Blackjack.

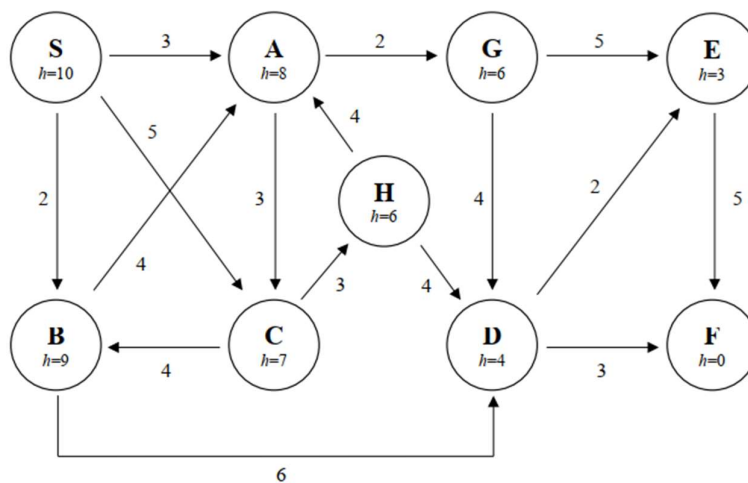
(<https://en.wikipedia.org/wiki/Blackjack>)

Please write your answer in the table

Property	Explanation
Partially observable	The game of Blackjack is partially observable since players can only see their own cards.
Multiagent	Blackjack can involve multiple players (agents) competing against each other and the dealer. At least 2 players (player and dealer).
Stochastic	The game of Blackjack involves an element of randomness due to the shuffling of the deck, making it stochastic. The outcome of drawing cards from the shuffled deck is probabilistic.
Sequential	Blackjack is a sequential game where the actions of each player, including the dealer, depend on the actions taken by the previous player(s).

Static	The game environment in Blackjack is static because the state of the game changes with each player's action. Players make decisions based on the evolving state of the game, such as the cards they receive.
Discrete	Players can choose from a limited set of actions (e.g., hit, stand, double down, split), and the possible states of the game (e.g., player hand value, dealer's visible card) are also discrete.
Known	In Blackjack, all players have complete information about the rules of the game, including the values assigned to different cards and the possible actions they can take.

Problem 5. (2.5pts) Consider the following graph. The start and goal states are S and F respectively.



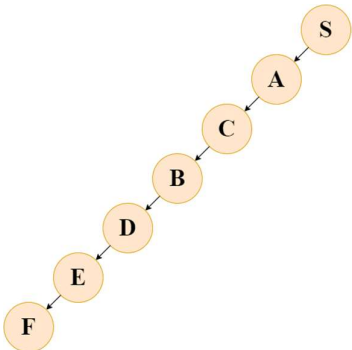
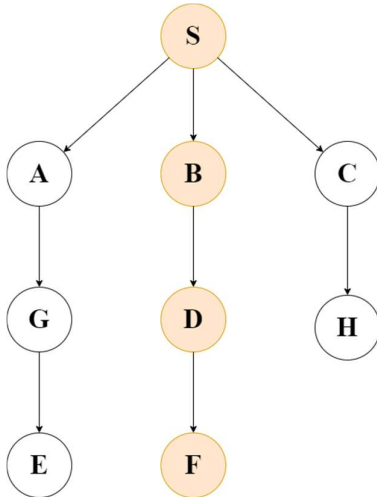
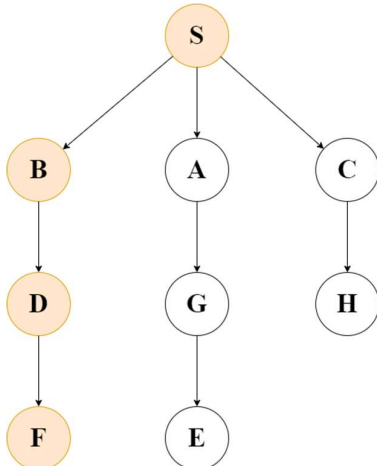
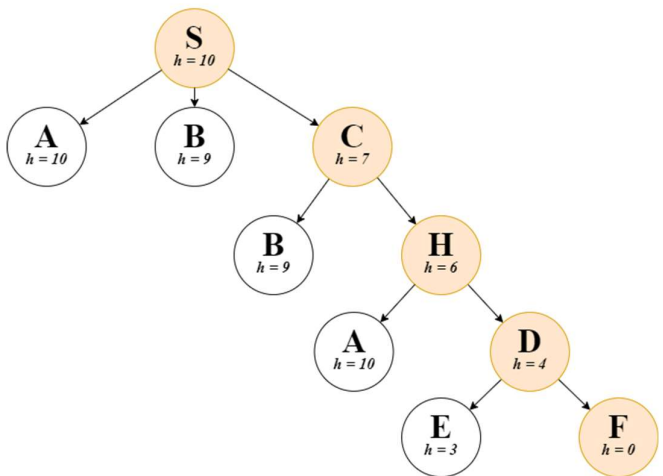
For each of the following graph search strategies, work out order in which states are expanded, as well as the path and show the search tree used to find this solution. In all cases, assume ties resolve in such a way that states with earlier alphabetical order are expanded first. Remember that in graph search, a state is expanded once.

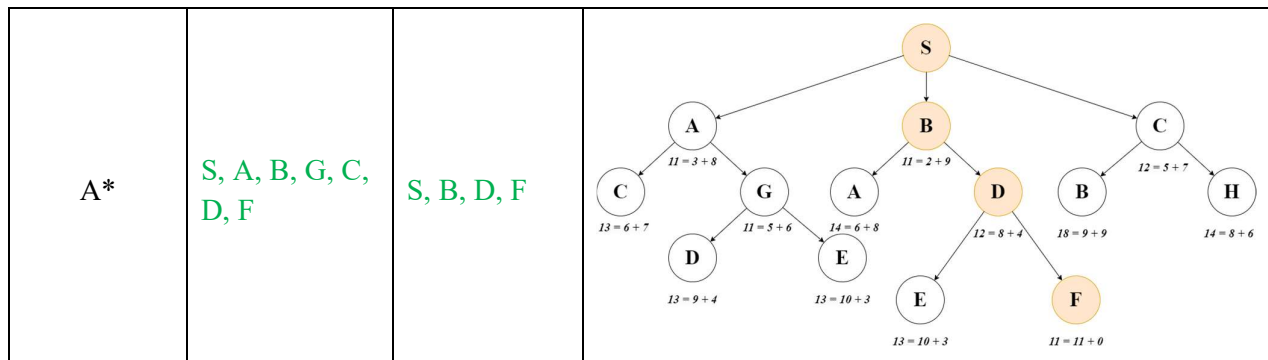
- Tree-search depth-first search (DFS)
- Breadth-first search (BFS)
- Uniform cost search (UCS)
- Greedy best first search (GBFS) with the heuristic h shown on the graph.
- Graph-search A* with the same heuristic.

Note that:

- A state is expanded at most once in graph search, while it may be expanded more than once in tree search.
- Tree-search DFS avoids repeated states by checking new states against those on the path from the root to the current node.
- For DFS, BFS, and GBFS, the goal test is applied to each node when it is generated rather than when it is selected for expansion.

Algorithms	List of expanded states (in exact order)	Path Returned	Search tree

DFS	S, A, C, B, D, E	S, A, C, B, D, E, F	
BFS	S, A, B, C, G, D, H, E	S, B, D, F	
UCS	S, B, A, C, G, D, H, E, F	S, B, D, F (cost = 11)	
GBFS	S, A, B, C, H, D, E	S, C, H, D, F	



Problem 6. (2pt) You are given the initial state (i) and the goal state (ii) of an 8-puzzle as shown below.

2	8	3
1	6	4
7		5

(i)

1	2	3
8		4
7	6	5

(ii)

a) Apply graph-search A* with heuristic h_1 = **the number of misplaced numbered tiles**. Ties are broken randomly.

- Draw the search tree that includes all states generated during the algorithm procedure.
- Compute the triple (g, h, f) for each state.
- Mark the optimal strategy found.

1	2	3	4	5	6	7	8	Total
X	X	✓	✓	✓	X	✓	X	4

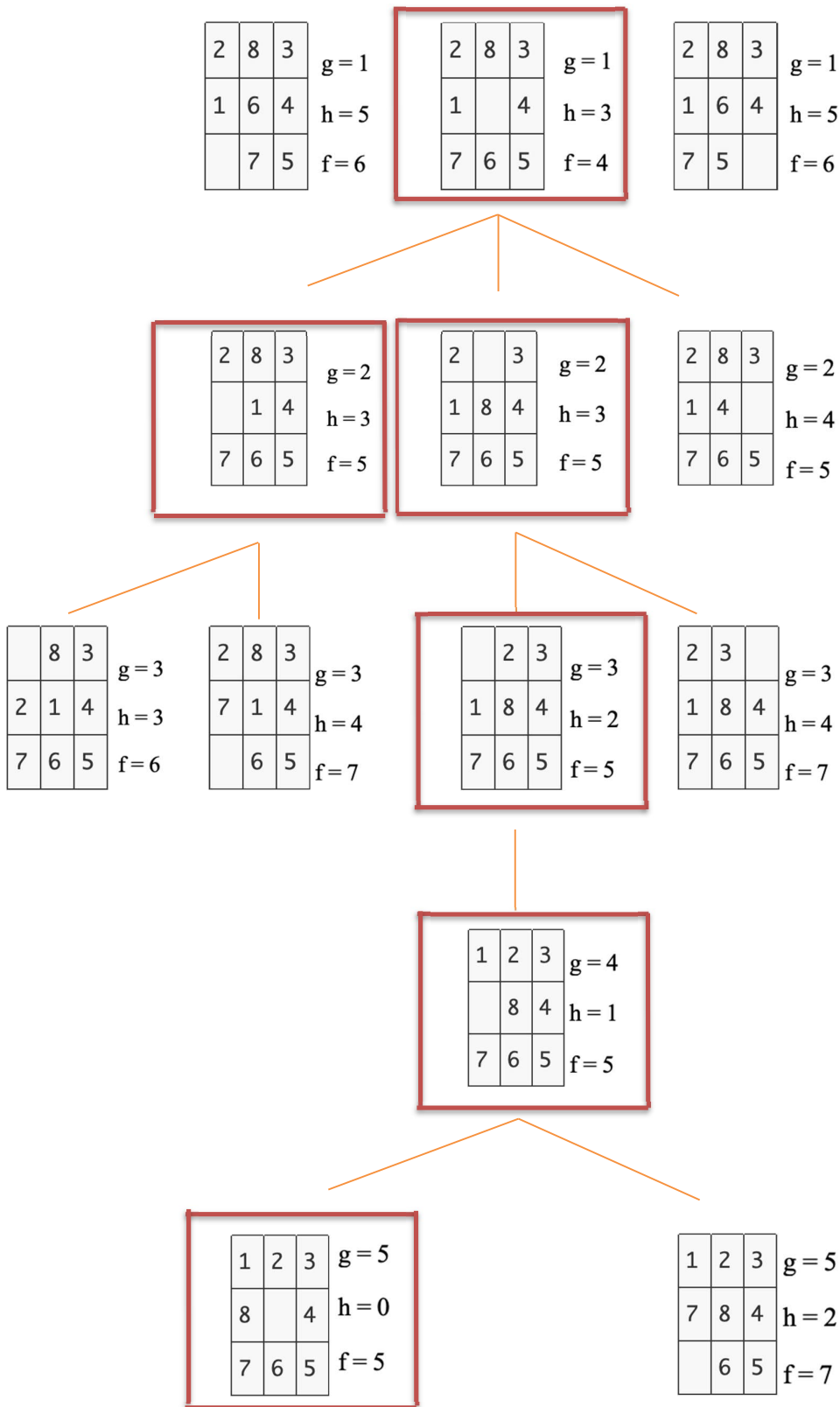
The search tree

2	8	3	g = 0
1	6	4	h = 4
7		5	f = 4



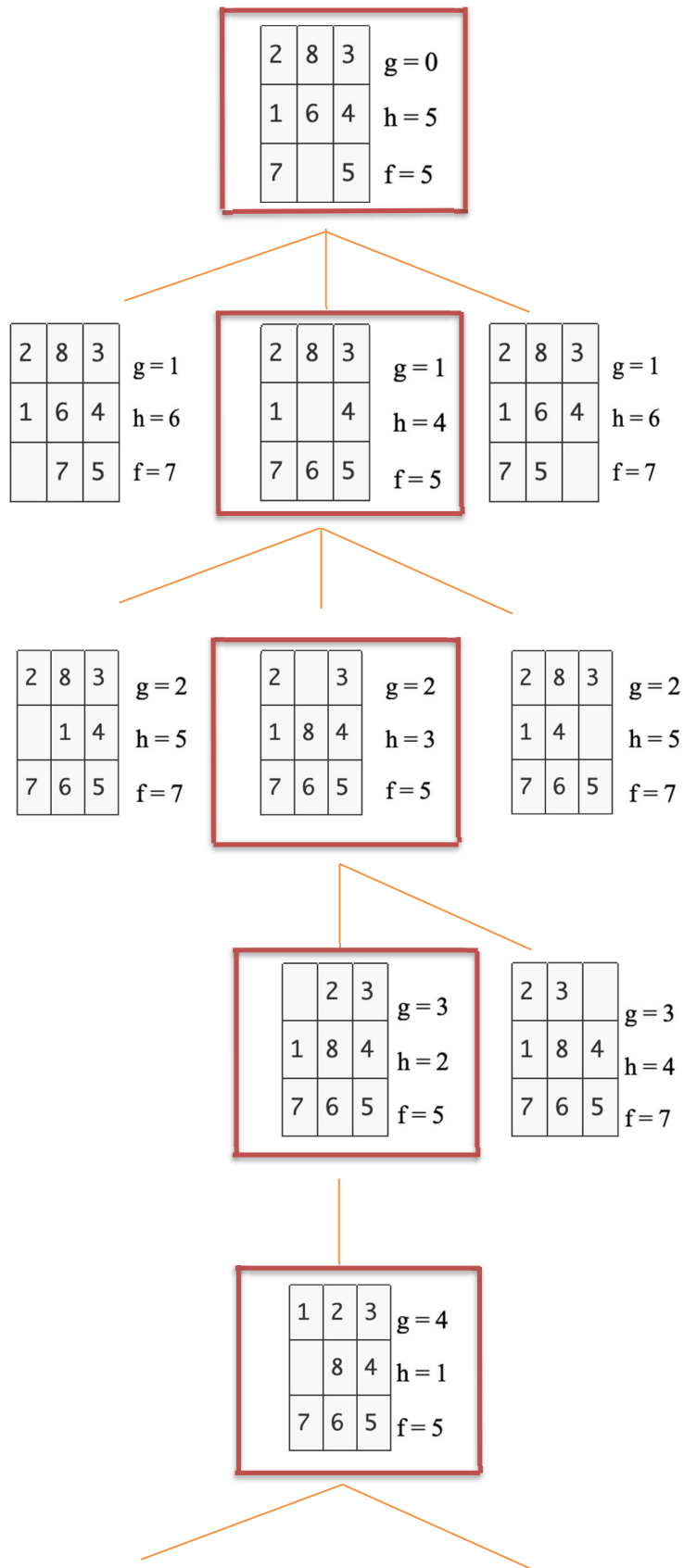
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b) Repeat the question a) with heuristic h_2 = the sum of the (Manhattan) distance of every numbered tile to its goal position.

The search tree



1	2	3	$g = 5$
8		4	$h = 0$
7	6	5	$f = 5$

1	2	3	$g = 5$
7	8	4	$h = 2$
	6	5	$f = 7$